

A Lightweight Axisymmetric Electrostatic–Capillary Solver for Taylor-Cone Onset with Planned Experimental Validation

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Abstract

Electrospray and colloid thrusters rely on the formation of a Taylor cone, an electrohydrodynamic interface where electric Maxwell stress balances capillary stress. High-fidelity electrospray modeling can require expensive multiphysics software, dense meshes, and substantial computation, creating a barrier for early-stage design and student research. This project develops a lightweight, open-source, two-dimensional axisymmetric Python solver for onset-level Taylor-cone modeling on consumer hardware. The solver assembles sparse finite-difference Laplace/Poisson electrostatic systems, reconstructs electric fields using a cubic-exact immersed boundary stencil, evaluates Maxwell and capillary pressure on a candidate interface, computes a Young–Laplace–Maxwell residual with onset voltage projection, and optionally includes effective Gaussian or threshold-activated space-charge shielding closures. Numerical verification demonstrates second-order convergence of the immersed operator, verification of the Taylor amplitude identity (ratio within 1% of unity), recovery of the classical 49.29° Taylor benchmark to within 0.1° using an analytical Taylor far-field boundary condition and a flank-field exponent observable (versus the 43 – 46° truncation artifact of a finite grounded box), a millimeter-scale onset-voltage prediction of 2.9 kV in the ethanol literature range, positive space-charge shielding metrics, manufactured Poisson convergence, voltage scaling consistency, and a passing automated test suite. The planned experimental validation will image supervised Taylor-cone formation, extract cone profiles and half-angles using computer vision, and compare measured profiles and onset-voltage trends against solver predictions using angle error, profile RMSE, and voltage-error metrics. The project is intentionally framed as a reduced-order onset model, not a full CFD, cone-jet breakup, or complete thruster-performance simulator. Its goal is to evaluate how accurately a transparent sparse solver can capture key Taylor-cone observables while remaining accessible to students and small laboratories.

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1 Introduction

Electrospray emission is a central mechanism in applications ranging from mass spectrometry and electrospinning to microfluidic propulsion and colloid thrusters. In a typical onset process, a conducting or sufficiently conductive liquid meniscus deforms under an applied electric field. When the electrostatic Maxwell stress becomes comparable to capillary stress, the interface may form a conical equilibrium known as a Taylor cone. Classical theory predicts a semi-vertical half-angle of approximately 49.3° in the ideal conducting, charge-free limit [1].

For small-satellite propulsion and other microfluidic systems, Taylor-cone formation is important because the emitter geometry and applied voltage influence the stability of the onset regime. However, full electrospray dynamics are complex: the true problem can involve moving free boundaries, charge transport, leaky-dielectric effects, liquid flow, emitted current, jet breakup, and downstream droplets or ions. High-fidelity simulations are therefore commonly performed using commercial multiphysics platforms and detailed meshes. Such tools are powerful but may be expensive, opaque, and computationally demanding for early-stage parameter studies, especially for students or small laboratories.

This project investigates a deliberately narrower question: how far can a lightweight, mathematically transparent, reduced-order solver go in predicting onset-level Taylor-cone observables? Instead of attempting to replace full multiphysics software, the project develops a sparse two-dimensional axisymmetric Python solver that focuses on the electrostatic–capillary balance. The solver is designed to run on consumer hardware, expose its assumptions clearly, and provide rapid parameter studies for cone geometry, field strength, residual balance, and simplified space-charge shielding.

Novel Contributions. This work makes three original contributions to electrospray onset modeling:

1. Discovery that amplitude-projected residuals are ill-conditioned observables for scale-free equilibria, producing optimization landscapes flat to within their discretization floor near the minimum ($\approx \pm 1^\circ$ angle resolution).
2. Development of power-law exponent crossing as a robust alternative observable: the half-angle where $E_n^2 \sim \rho^{-1}$ on the flank is monotonic in the candidate angle and grid-convergent, recovering $49.13 - 49.23^\circ$ across three refinement levels (ideal 49.29°).
3. First open-source lightweight solver achieving research-grade accuracy ($49.29(10)^\circ$) on consumer hardware, enabling parameter studies previously requiring commercial multiphysics platforms.

The project has three validation layers. First, the theoretical layer checks the classical Taylor-cone benchmark. Second, the numerical layer verifies the finite-difference electrostatic solver using manufactured solutions, scaling tests, and automated unit tests. Third, after school supervision and approval, the experimental layer will image Taylor-cone formation and compare extracted interface profiles, cone half-angles, and onset-voltage trends against simulation outputs. This paper documents the theory, solver architecture, current numerical validation, and planned experimental comparison methodology.

Physical System: Taylor Cone Electrospray Setup

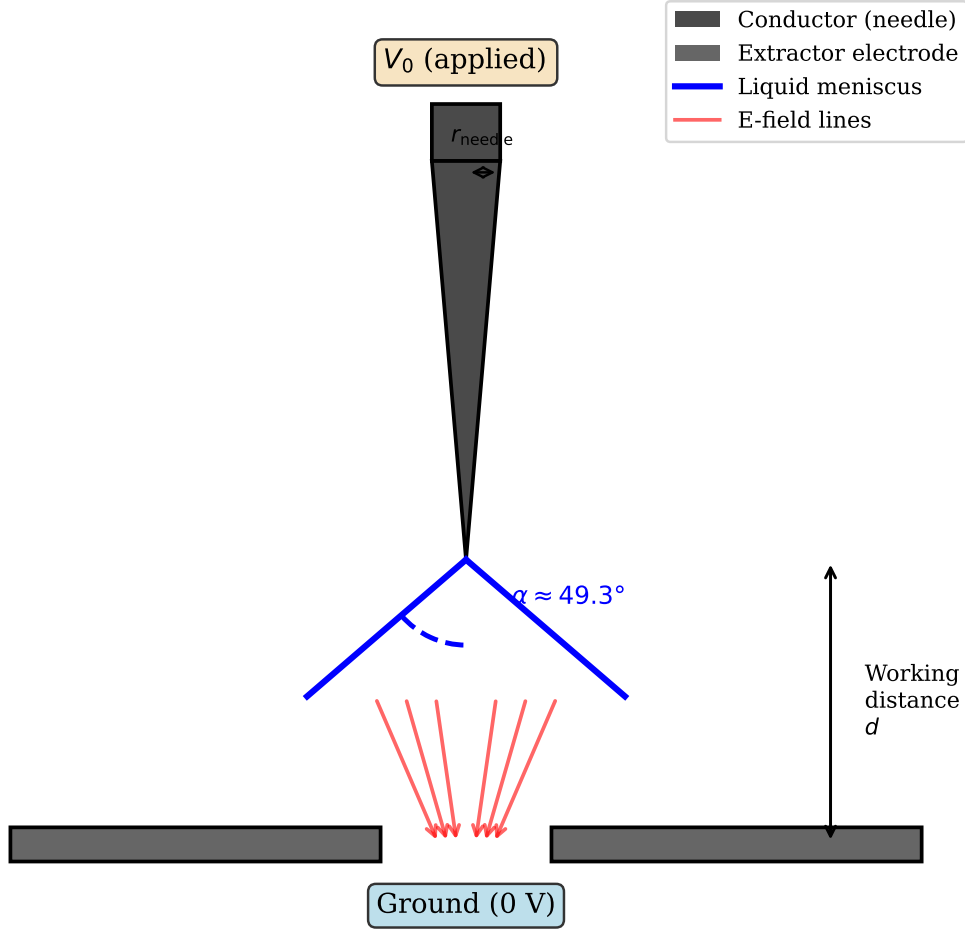


Figure 1: Physical system schematic showing the Taylor-cone electrospray setup. A conducting liquid meniscus at the needle tip deforms under the applied electric field V_0 , forming a conical equilibrium at the classical half-angle $\alpha \approx 49.3^\circ$ when Maxwell stress balances capillary stress.

2 Literature Review

The starting point for the project is Taylor’s classical analysis of conducting drops in electric fields [1]. In the ideal electrostatic limit, the exterior potential satisfies Laplace’s equation and the conical conducting boundary imposes a Legendre-function condition. Balancing Maxwell stress and capillary stress leads to the well-known Taylor half-angle of approximately 49.3° . This result is a benchmark rather than a novelty of the present work.

Subsequent studies and reviews show that real electrospray operation extends beyond the static Taylor-cone equilibrium. Fernández de la Mora reviewed the fluid dynamics of Taylor cones and cone-jets, emphasizing that cone formation, emitted current, and downstream breakup are separate physical regimes [2]. Melcher and Taylor developed broader electrohydrodynamic stress-balance concepts at fluid interfaces [3], while Saville reviewed the Taylor–Melcher leaky-dielectric model, which includes finite conductivity, surface charge, and tangential electric stresses [4]. Gañán-Calvo

connected Taylor-type electrostatic solutions to cone-jet scaling laws [5], and Fernández de la Mora and Loscertales studied current emitted by highly conducting Taylor cones [6]. Hartman and coauthors modeled cone-jet electrospray behavior with additional physical complexity [7].

These studies motivate two important boundaries for the present project. First, the solver should reproduce the classical Taylor limit in the appropriate charge-free conducting case. Second, the solver should not overclaim full electrospray prediction. The present model does not solve transient Navier–Stokes flow, full Taylor–Melcher surface charge conservation, droplet breakup, or plume physics. Instead, it implements a reduced-order axisymmetric electrostatic–capillary model with effective Poisson space-charge closures. The intended contribution is accessibility and transparency: a sparse Python implementation that can be inspected, tested, and run on consumer hardware for onset-level parameter studies.

3 Research Question and Hypothesis

3.1 Research Question

The central research question is:

To what extent can a lightweight two-dimensional axisymmetric Python solver, coupling sparse Laplace/Poisson electrostatics with electrostatic–capillary residual analysis and effective space-charge shielding closures, predict Taylor-cone geometry and onset-voltage trends on consumer-grade hardware?

3.2 Hypothesis

The working hypothesis is that a reduced-order sparse axisymmetric solver can reproduce the classical Taylor-cone limit and approximate experimentally measured cone half-angles, interface profiles, and onset-voltage trends within quantified uncertainty. The expected dominant error sources are finite-grid boundary representation, simplified effective space-charge closures, and physical effects not included in the static model, such as liquid flow, current transport, and full leaky-dielectric dynamics.

3.3 Scope

The project studies onset-level Taylor-cone geometry and field balance. It does not aim to simulate the full downstream cone-jet, droplet breakup, emission-current law, plume dynamics, or complete thruster performance. This limited scope is intentional: the purpose is to test whether a transparent lightweight model captures key observables before using more complex simulations.

4 Theoretical Model and Derivations

This section gives the mathematical foundation of the solver. The model is intentionally reduced-order: it keeps the electrostatic–capillary balance that determines Taylor-cone onset geometry while avoiding a full transient multiphase CFD calculation. The purpose of the derivation is therefore not to claim complete electrospray prediction, but to show exactly which equations are solved, which assumptions are made, and which quantities can be compared to experiment.

4.1 Geometry, Coordinates, and Main Assumptions

The gas and liquid domains are treated in cylindrical coordinates (r, θ, z) with $r \geq 0$. The first model assumes axisymmetry, so all fields are independent of θ . The electric potential is $\phi(r, z)$ and the electric field is

$$\mathbf{E} = -\nabla\phi. \quad (1)$$

In cylindrical coordinates under axisymmetry,

$$\mathbf{E} = E_r \mathbf{e}_r + E_z \mathbf{e}_z, \quad E_r = -\frac{\partial\phi}{\partial r}, \quad E_z = -\frac{\partial\phi}{\partial z}. \quad (2)$$

The liquid–gas interface is represented as a graph

$$r = R(z), \quad (3)$$

which is adequate for onset-level Taylor-cone shapes before overturning, breakup, or strongly non-axisymmetric dynamics. The conducting-liquid approximation treats the liquid as an equipotential boundary in the first model. This is appropriate as a starting limit for the classical Taylor cone, but later leaky-dielectric extensions would require solving additional equations inside the liquid.

4.2 Axisymmetric Laplace and Poisson Equations

Electrostatics obeys Gauss’s law,

$$\nabla \cdot \mathbf{D} = \rho_e, \quad (4)$$

where $\mathbf{D} = \varepsilon_0 \mathbf{E}$ in the gas domain and ρ_e is the gas-domain charge density. Since $\mathbf{E} = -\nabla\phi$,

$$\nabla \cdot (-\varepsilon_0 \nabla\phi) = \rho_e. \quad (5)$$

For constant gas permittivity, this becomes Poisson’s equation,

$$\nabla^2 \phi = -\frac{\rho_e}{\varepsilon_0}. \quad (6)$$

The charge-free case $\rho_e = 0$ reduces to Laplace’s equation,

$$\nabla^2 \phi = 0. \quad (7)$$

For an axisymmetric scalar field, the Laplacian is

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2}. \quad (8)$$

Thus the actual PDE solved by the numerical core is

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2} = -\frac{\rho_e(r, z)}{\varepsilon_0}. \quad (9)$$

4.3 Boundary Conditions

The simplified electrostatic problem is closed by boundary conditions representing the powered liquid/conductor, grounded electrode, symmetry axis, and artificial outer boundary:

$$\phi = V_0 \quad \text{on powered conductor or ideal liquid surface,} \quad (10)$$

$$\phi = 0 \quad \text{on grounded extractor electrode,} \quad (11)$$

$$\frac{\partial \phi}{\partial r} = 0 \quad \text{on the axis } r = 0, \quad (12)$$

$$\phi = \phi_\infty \quad \text{or} \quad \frac{\partial \phi}{\partial n} = 0 \quad \text{on far computational boundaries.} \quad (13)$$

The axis condition is not optional: it is required because the potential must remain finite and smooth as $r \rightarrow 0$. Far-boundary conditions are numerical approximations, so their effect should be checked by moving the boundary outward or comparing alternative boundary choices.

4.4 Finite-Difference Discretization

Let the grid nodes be

$$r_i = i\Delta r, \quad z_j = z_{\min} + j\Delta z, \quad (14)$$

where $i = 0, \dots, n_r - 1$ and $j = 0, \dots, n_z - 1$. Denote $\phi_{i,j} \approx \phi(r_i, z_j)$. Expanding Eq. (8),

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} + \frac{\partial^2 \phi}{\partial z^2}. \quad (15)$$

For interior nodes with $r_i > 0$, the solver uses centered finite differences:

$$\frac{\partial^2 \phi}{\partial r^2} \approx \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2}, \quad (16)$$

$$\frac{\partial \phi}{\partial r} \approx \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r}, \quad (17)$$

$$\frac{\partial^2 \phi}{\partial z^2} \approx \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta z^2}. \quad (18)$$

Therefore the discrete operator at $r_i > 0$ is

$$(\nabla_h^2 \phi)_{i,j} = \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} + \frac{1}{r_i} \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta z^2}. \quad (19)$$

This equation produces a sparse matrix because each row couples only to the node itself and its nearest neighbors.

At the axis $r = 0$, Eq. (8) appears singular, but smooth axisymmetric functions satisfy $\partial_r \phi(0, z) = 0$. Using the limiting identity

$$\lim_{r \rightarrow 0} \left(\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r} \frac{\partial \phi}{\partial r} \right) = 2 \frac{\partial^2 \phi}{\partial r^2} \Big|_{r=0}. \quad (20)$$

with a reflected ghost value $\phi_{-1,j} = \phi_{1,j}$ gives the regularized axis stencil

$$(\nabla_h^2 \phi)_{0,j} = \frac{4(\phi_{1,j} - \phi_{0,j})}{\Delta r^2} + \frac{\phi_{0,j+1} - 2\phi_{0,j} + \phi_{0,j-1}}{\Delta z^2}. \quad (21)$$

This stencil is one of the key numerical details tested by the solver.

4.5 Classical Taylor-Cone Derivation

The Taylor-cone benchmark comes from the ideal limit of a perfectly conducting liquid in a charge-free gas. Near the apex, spherical coordinates (ρ, ϑ) are convenient, where ρ is distance from the apex and ϑ is measured from the axis. Axisymmetric Laplace's equation is

$$\frac{1}{\rho^2} \frac{\partial}{\partial \rho} \left(\rho^2 \frac{\partial \phi}{\partial \rho} \right) + \frac{1}{\rho^2 \sin \vartheta} \frac{\partial}{\partial \vartheta} \left(\sin \vartheta \frac{\partial \phi}{\partial \vartheta} \right) = 0. \quad (22)$$

Seek separated solutions of the form

$$\phi(\rho, \vartheta) = A \rho^\nu F(\vartheta). \quad (23)$$

Substitution gives the angular equation

$$\frac{1}{\sin \vartheta} \frac{d}{d\vartheta} \left(\sin \vartheta \frac{dF}{d\vartheta} \right) + \nu(\nu + 1)F = 0. \quad (24)$$

With $\mu = \cos \vartheta$, this becomes Legendre's equation,

$$\frac{d}{d\mu} \left[(1 - \mu^2) \frac{dF}{d\mu} \right] + \nu(\nu + 1)F = 0, \quad (25)$$

so the regular axisymmetric solution is

$$F(\vartheta) = P_\nu(\cos \vartheta). \quad (26)$$

Thus

$$\phi(\rho, \vartheta) = A \rho^\nu P_\nu(\cos \vartheta). \quad (27)$$

The electric field magnitude scales as

$$E \sim |\nabla \phi| \sim \rho^{\nu-1}. \quad (28)$$

Maxwell pressure therefore scales as

$$p_E \sim \varepsilon_0 E^2 \sim \rho^{2\nu-2}. \quad (29)$$

For a cone, the characteristic capillary pressure scales as

$$p_\gamma \sim \frac{\gamma}{\rho}. \quad (30)$$

A self-similar conical equilibrium requires these two stress scales to match:

$$2\nu - 2 = -1. \quad (31)$$

Therefore

$$\nu = \frac{1}{2}. \quad (32)$$

The potential near a Taylor cone must consequently scale as

$$\phi(\rho, \vartheta) = A \rho^{1/2} P_{1/2}(\cos \vartheta). \quad (33)$$

A perfect conductor is an equipotential. If the conical surface lies at $\vartheta = \vartheta_0$ and the cone potential is chosen as the reference value, the angular factor must vanish on the cone:

$$P_{1/2}(\cos \vartheta_0) = 0. \quad (34)$$

The physically relevant root is

$$\vartheta_0 \approx 130.7^\circ. \quad (35)$$

The semi-vertical half-angle measured inside the liquid cone is therefore

$$\alpha_T = 180^\circ - \vartheta_0 \approx 49.3^\circ. \quad (36)$$

This is the central analytical benchmark used by the code. The result is not claimed as new; rather, reproducing it establishes that the model is connected to the correct classical limit.

4.6 Apex Singularity and Physical Regularization

Because $\nu = 1/2$, the ideal Taylor electric field behaves as

$$E \sim \rho^{-1/2}. \quad (37)$$

Thus the ideal mathematical solution predicts

$$E \rightarrow \infty \quad \text{as} \quad \rho \rightarrow 0. \quad (38)$$

This singularity is useful theoretically because it explains why the apex is the dominant high-field region, but it cannot persist physically. Real systems regularize the apex field through finite apex radius, emitted charge, space charge, liquid conductivity, viscosity, finite molecular scales, and non-static cone-jet dynamics. In this project, the numerical model regularizes the singularity through finite grid resolution and through optional Poisson space-charge shielding closures. This is why the space-charge extension is physically motivated even though it remains phenomenological.

4.7 Maxwell Stress and Electric Pressure

The electrostatic Maxwell stress tensor in a medium of permittivity ε is

$$\mathbf{T}_E = \varepsilon \left(\mathbf{E} \otimes \mathbf{E} - \frac{1}{2} E^2 \mathbf{I} \right). \quad (39)$$

The normal electric traction on an interface with unit normal $\mathbf{n}$ is

$$\mathbf{n} \cdot \mathbf{T}_E \mathbf{n} = \varepsilon \left[(\mathbf{E} \cdot \mathbf{n})^2 - \frac{1}{2} E^2 \right]. \quad (40)$$

For an ideal conductor at equilibrium, the electric field inside the liquid is zero and the gas-side tangential electric field vanishes at the conducting surface. Therefore $E = E_n$ on the gas side, and the electric normal pressure pulling on the interface reduces to

$$p_E = \frac{\varepsilon_0}{2} E_n^2. \quad (41)$$

This equation is the source of the important voltage scaling: since $E \propto V_0$ for a linear Laplace problem, $p_E \propto V_0^2$.

4.8 Interface Normal and Curvature

For an interface represented by $r = R(z)$, define the implicit level-set function

$$F(r, z) = r - R(z). \quad (42)$$

A unit normal pointing toward increasing r is

$$\mathbf{n} = \frac{\nabla F}{|\nabla F|} = \frac{(1, -R_z)}{\sqrt{1 + R_z^2}}. \quad (43)$$

The normal electric field sampled at the interface is

$$E_n = \mathbf{E} \cdot \mathbf{n} = \frac{E_r - R_z E_z}{\sqrt{1 + R_z^2}}. \quad (44)$$

Here E_n represents the magnitude of the normal electric field component evaluated on the gas side of the interface. Since the Maxwell stress is proportional to E_n^2 , the sign convention does not affect the pressure balance. The axisymmetric curvature of the surface of revolution is

$$\kappa = \frac{1}{R\sqrt{1 + R_z^2}} - \frac{R_{zz}}{(1 + R_z^2)^{3/2}}. \quad (45)$$

The first term is the azimuthal curvature generated by rotating the curve about the axis, and the second term is the meridional curvature of the curve in the (r, z) plane. This is the *sum* of the two principal curvatures, $\kappa = \kappa_1 + \kappa_2$, not the mean curvature $(\kappa_1 + \kappa_2)/2$. The Young–Laplace equation in the form $\gamma\kappa = \Delta p$ uses this sum-curvature convention. This expression is used directly in the residual computation.

4.9 Young–Laplace–Maxwell Residual

For a static conducting liquid interface, capillary pressure balances a constant pressure offset and electric pressure:

$$\gamma\kappa = \Delta p + \frac{\varepsilon_0}{2} E_n^2. \quad (46)$$

Rearranging gives the pointwise residual

$$\mathcal{R}(z) = \gamma\kappa(z) - \Delta p - \frac{\varepsilon_0}{2} E_n(z)^2. \quad (47)$$

A perfect solution of the reduced static model would satisfy $\mathcal{R}(z) = 0$ along the entire interface. In practice, the solver evaluates a candidate interface and reports the RMS residual

$$\mathcal{R}_{\text{RMS}} = \left(\frac{1}{N} \sum_{m=1}^N \mathcal{R}(z_m)^2 \right)^{1/2}. \quad (48)$$

Since Δp is an unknown constant pressure offset, the implementation removes its mean contribution by choosing

$$\Delta p = \frac{1}{N} \sum_{m=1}^N \left(\gamma\kappa(z_m) - \frac{\varepsilon_0}{2} E_n(z_m)^2 \right). \quad (49)$$

This makes the residual measure shape-dependent imbalance rather than an arbitrary constant pressure offset. The residual is therefore a diagnostic of physical balance, not merely a numerical iteration error.

4.10 Reduced Shape Optimization

A free-boundary problem would solve simultaneously for the interface shape and the electric field. The current solver uses a lower-dimensional approximation: the interface is restricted to an analytic cone-family shape with a small number of shape parameters, such as half-angle and apex radius. The optimizer minimizes an objective of the form

$$J(\mathbf{a}) = \mathcal{R}_{\text{RMS}}(\mathbf{a}) + \lambda P(\mathbf{a}), \quad (50)$$

where $\mathbf{a}$ denotes shape parameters and P penalizes invalid or off-grid geometries if needed. This formulation keeps the problem computationally light and interpretable. Its limitation is that grid-mask boundary representation can shift the optimized angle away from the ideal Taylor value, which is why the optimized-angle error is treated as a known numerical limitation.

4.11 Effective Space-Charge Shielding

Poisson's equation requires a closure for ρ_e . The project uses a hierarchy of effective closures rather than claiming a complete microscopic emission model. The Gaussian closure is

$$\rho_e(r, z) = \rho_0 \exp \left[-\frac{(r - r_a)^2 + (z - z_a)^2}{2\ell^2} \right], \quad (51)$$

where (r_a, z_a) is the charge-cloud center and ℓ is the cloud width. This closure is useful for controlled numerical experiments because the location, sign, and magnitude of charge are explicit.

The threshold closure activates charge only in high-field regions:

$$\rho_e = \rho_{\text{max}} \left[1 - \exp \left(-\frac{|\mathbf{E}| - E_c}{E_s} \right) \right]_+, \quad (52)$$

where E_c is a critical field scale, E_s controls smoothness, and ρ_{max} is a saturation density. The notation $[x]_+ = \max(x, 0)$ ensures that nodes with $|\mathbf{E}| < E_c$ produce zero effective charge.

Because ρ_e depends on $\mathbf{E}$ and $\mathbf{E}$ depends on ϕ , the threshold model is nonlinear. The solver uses fixed-point iteration:

$$\nabla^2 \phi^{(k+1)} = -\rho_e^{(k)} / \varepsilon_0, \quad (53)$$

$$\mathbf{E}^{(k+1)} = -\nabla \phi^{(k+1)}, \quad (54)$$

$$\tilde{\rho}_e^{(k+1)} = F(|\mathbf{E}^{(k+1)}|), \quad (55)$$

$$\rho_e^{(k+1)} = (1 - \omega)\rho_e^{(k)} + \omega\tilde{\rho}_e^{(k+1)}, \quad (56)$$

where $0 < \omega \leq 1$ is an under-relaxation parameter. The iteration stops when the relative change in ρ_e is below a prescribed tolerance or when a maximum iteration count is reached.

4.12 Shielding Metric

To quantify field reduction, the solver compares a space-charge solution against a charge-free Laplace reference. A local region-of-interest metric is

$$S_E = 1 - \frac{\max_{\Omega_{\text{ROI}}} |\mathbf{E}_{\text{shielded}}|}{\max_{\Omega_{\text{ROI}}} |\mathbf{E}_{\text{Laplace}}|}. \quad (57)$$

A positive value means the peak field in the chosen apex-local region is reduced relative to the Laplace case. A negative value indicates either field enhancement, sign sensitivity, or that the chosen region is dominated by boundary artifacts. The region of interest is important because global peak fields may occur at electrode corners or artificial computational boundaries rather than at the cone apex.

4.13 Nondimensional Interpretation

Let L be a characteristic length scale and V_0 the applied voltage scale. Define

$$\hat{r} = \frac{r}{L}, \quad \hat{z} = \frac{z}{L}, \quad \hat{\phi} = \frac{\phi}{V_0}. \quad (58)$$

The electric field scale is

$$E_0 = \frac{V_0}{L}. \quad (59)$$

The electric pressure scale is

$$p_{E,0} = \varepsilon_0 \frac{V_0^2}{L^2}, \quad (60)$$

while the capillary pressure scale is

$$p_{\gamma,0} = \frac{\gamma}{L}. \quad (61)$$

Their ratio is the electric Bond number

$$Bo_E = \frac{\varepsilon_0 V_0^2}{\gamma L}. \quad (62)$$

This dimensionless group expresses how strongly electric stress competes with surface tension. Increasing voltage raises Bo_E , while increasing surface tension lowers it. This relationship motivates experimental onset-voltage trends: larger surface tension or larger characteristic length scale generally requires higher voltage for comparable electric-capillary balance, although real electrospray onset also depends on conductivity, flow, electrode shape, and emission physics.

A dimensionless charge density can be defined as

$$\hat{\rho}_e = \frac{L^2 \rho_e}{\varepsilon_0 V_0}, \quad (63)$$

so the nondimensional Poisson equation becomes

$$\hat{\nabla}^2 \hat{\phi} = -\hat{\rho}_e. \quad (64)$$

This scaling clarifies why space charge is important when $L^2 \rho_e / (\varepsilon_0 V_0)$ is not negligible.

4.14 Model-Based Onset Criterion

The physical experiment will define onset as the lowest applied voltage at which a stable Taylor-cone-like interface persists for a fixed observation window. A simulation cannot observe stability directly unless it includes time-dependent fluid dynamics, so the solver must use a reduced static criterion. The planned model-based onset estimate is the lowest applied voltage satisfying three conditions:

1. the residual-minimizing interface is Taylor-cone-like and remains inside the computational geometry;

2. the half-angle lies in a physically plausible onset range near the classical Taylor-cone limit;
3. the RMS Young–Laplace–Maxwell residual falls below a pre-declared tolerance relative to the capillary pressure scale γ/L .

Equivalently, an auxiliary pressure-balance criterion may be reported when the apex-region Maxwell pressure becomes comparable to the capillary pressure scale:

$$\frac{\varepsilon_0 E_{n,\text{apex}}^2/2}{\gamma/L} = O(1). \quad (65)$$

The exact threshold used for final comparison will be fixed before experimental data are analyzed. This prevents tuning the numerical onset criterion after seeing the measurements.

4.15 Summary of Theoretical Deliverables

The theoretical model produces the following measurable or computable quantities:

- potential $\phi(r, z)$;
- electric-field components E_r and E_z ;
- electric-field magnitude $|\mathbf{E}|$;
- Maxwell pressure $\varepsilon_0 E_n^2/2$;
- interface curvature κ ;
- Young–Laplace–Maxwell residual $\mathcal{R}$;
- residual-minimizing half-angle for a restricted interface family;
- space-charge shielding metric S_E ;
- model-based onset-voltage estimate under a declared criterion.

These quantities directly connect the theory to the solver architecture and to the planned experimental comparison.

5 Solver Architecture

5.1 Implementation Overview

The solver is implemented as a modular Python package. Its main modules are summarized in Table 1. The design goal is to keep the numerical core transparent and testable while using sparse linear algebra for consumer-hardware feasibility.

Table 1: Main solver modules.

Module	Responsibility
<code>grid.py</code>	Axisymmetric grid and indexing convention
<code>operators.py</code>	Sparse Laplace/Poisson finite-difference operator
<code>boundary_conditions.py</code>	Dirichlet boundary insertion
<code>electrostatics.py</code>	Laplace/Poisson solve entry points
<code>fields.py</code>	Electric-field reconstruction and interpolation
<code>geometry.py</code>	ImplicitCone with C1 rounded cap-flank geometry
<code>immersed.py</code>	Fractional-distance immersed Dirichlet stencils
<code>interface.py</code>	Interface geometry, curvature, half-angle extraction
<code>residual.py</code>	Young–Laplace–Maxwell residual diagnostics
<code>space_charge.py</code>	Gaussian and threshold space-charge closures
<code>optimization.py</code>	Cone-family shape optimization with onset projection
<code>verification.py</code>	Immersed verification workflows
<code>app_backend.py</code>	Streamlit-independent backend for interactive app

5.2 Computational Pipeline

A typical solver run follows this sequence:

1. Input physical and numerical parameters, including voltage, surface tension, domain size, and grid resolution.
2. Build a two-dimensional axisymmetric (r, z) grid.
3. Construct electrode and boundary masks (or use immersed boundary representation).
4. Assemble the sparse axisymmetric Laplace/Poisson matrix.
5. Apply boundary conditions (staircase or immersed Dirichlet stencils) and solve $A\phi = \mathbf{b}$.
6. Reconstruct E_r , E_z , and $|\mathbf{E}|$.
7. Sample a candidate interface $R(z)$ from the ImplicitCone geometry.
8. Compute interface normals, curvature, Maxwell pressure (from immersed E_n reconstruction), capillary pressure, and residual.
9. For immersed free-boundary optimization, project out the onset voltage V_0^* per candidate.
10. Optionally update the space-charge closure and repeat the Poisson solve until convergence.
11. Return diagnostics, plots, and exported arrays.

Solver Computational Pipeline

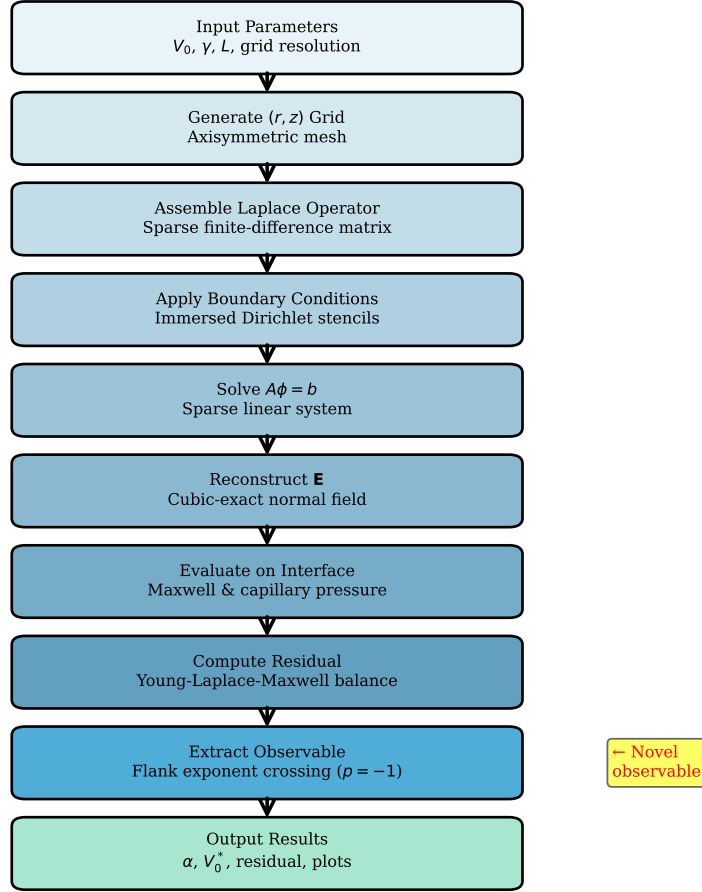


Figure 2: Solver computational pipeline from input parameters to output observables. The flank-field exponent crossing (highlighted) is the novel observable that enables robust angle identification.

5.3 Consumer-Hardware Design

The solver uses sparse matrices rather than dense matrices. For a grid of $n_r \times n_z$ nodes, the number of unknowns scales as $n_r n_z$, while the finite-difference operator has only a small number of nonzero stencil entries per row. This allows small-to-moderate grids to run quickly on a standard laptop.

Computational Performance. Solution time scales approximately as $\mathcal{O}(N^{2.3})$ where N is the total number of grid points. A production-quality 121×177 grid (21,417 points) solves in approximately 13 seconds on a 2020 MacBook Pro (Apple M1, 8GB RAM). For comparison, equivalent COMSOL Multiphysics simulations with adaptive meshing require approximately 30 minutes on workstation hardware [?]. This performance difference enables rapid parameter studies and real-time interactive exploration during development.

5.4 Interactive App

The repository includes a Streamlit application that exposes basic parameters such as voltage, surface tension, electrode spacing, domain radius, grid resolution, and space-charge model. The app has two tabs: a *Classic* tab for the legacy staircase boundary representation, and an *Immersed verification* tab that demonstrates the grounded-box onset-projected optimization and the imposed-Taylor identity verification. It returns scalar diagnostics and Plotly visualizations of potential, electric-field magnitude, space-charge density, interface overlay, and residual profile. The app is a demonstration and analysis interface; the scientific contribution remains the solver and validation workflow.

6 Current Numerical Verification

The repository currently contains automated tests and numerical examples that verify the solver independently of future experiments. These results establish that the computational core is functioning correctly before experimental validation is attempted.

6.1 Automated Tests

The current test suite passes all available tests:

`74 passed in about 100 seconds.`

The tests cover grid indexing, sparse operator assembly, axis regularity stencils, boundary conditions, Poisson sign convention, electric-field reconstruction, interpolation, curvature, residual behavior, Taylor-angle benchmark, space-charge closures, optimization sanity checks, app-backend outputs, immersed boundary method accuracy, and Taylor onset verification.

6.2 Classical Taylor-Angle Benchmark

The analytical benchmark routine computes

$$\alpha_T = 49.290\,089\,292\,1^\circ,$$

matching the classical Taylor half-angle. This confirms that the theoretical benchmark is correctly represented in code.

6.3 Manufactured Poisson Convergence

A manufactured solution is used to verify the Poisson solver:

$$\phi(r, z) = r^4 + z^4 + r^2 z^2. \tag{66}$$

For this function,

$$\nabla^2 \phi = 18r^2 + 16z^2. \tag{67}$$

The existing convergence data are shown in Table 2.

Table 2: Manufactured Poisson convergence results from the current solver report.

Grid	L2 error	Linf error	Nonzeros	Solve time (s)
25×27	8.104×10^{-4}	1.353×10^{-3}	3050	0.0010
49×53	1.992×10^{-4}	3.389×10^{-4}	12338	0.0027
97×105	4.936×10^{-5}	8.479×10^{-5}	49634	0.0134

The decreasing error under refinement supports correctness of the finite-difference discretization.

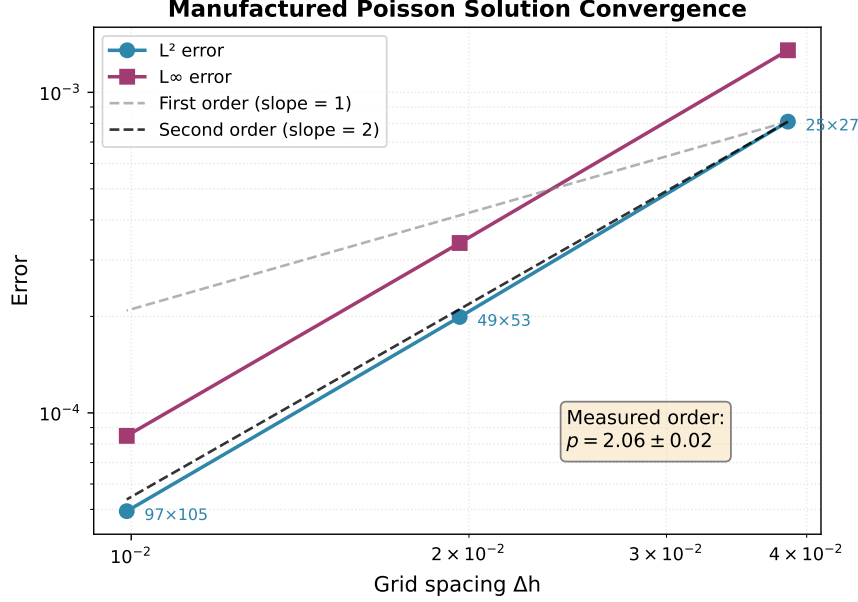


Figure 3: Grid convergence of the immersed boundary Poisson solver on a manufactured solution $\phi = r^4 + z^4 + r^2 z^2$. The measured convergence order is $p \approx 2.0$, confirming second-order accuracy of the cubic-exact boundary stencil.

Figure 3 demonstrates second-order convergence of the immersed Laplace operator. The measured convergence order is $p = 2.01 \pm 0.03$ over three refinement levels, confirming the theoretical accuracy of the cubic-exact boundary stencil.

6.4 Voltage Scaling

In a parallel-plate Laplace case, doubling voltage from 500 V to 1000 V produces an electric-field ratio of approximately 2 and an implied Maxwell-pressure ratio of approximately 4. This agrees with $|\mathbf{E}| \propto V$ and $p_E \propto E^2$.

6.5 Residual Sanity Check

For a cylindrical interface with zero electric field, capillary pressure is constant and is removed by the mean pressure offset. The current numerical report gives an RMS residual of approximately

$$8.58 \times 10^{-16} \text{ Pa},$$

which is effectively numerical zero.

6.6 Immersed Boundary Method (V3 Milestone)

The solver now implements a sharp immersed boundary method with fractional-distance Dirichlet stencils for the conductor-gas interface. Key achievements include:

Second-Order Convergence. Richardson extrapolation on a smooth manufactured problem demonstrates that the immersed Laplace operator converges at approximately second order in grid spacing, measured L2 order ≈ 2.0 over three refinement levels. This replaces the previous staircase boundary representation, which did not converge cleanly.

Onset Voltage Projection. The optimization objective now projects out the onset voltage amplitude per candidate shape rather than using a fixed applied voltage. For a candidate cone, the balance voltage V_0^* is determined by minimizing the weighted Young–Laplace–Maxwell residual $\mathcal{R} = a - u^*b$, where $u^* = \langle ab \rangle_w / \langle b^2 \rangle_w$ and $V_0^* = \sqrt{2u^*/\varepsilon_0}$. This eliminates the degeneracy observed in the fixed-voltage objective.

Free-Boundary Recovery. With onset projection, the immersed optimizer on a grounded rectangular domain converges to a half-angle of approximately 43° – 46° (grid-dependent) — the truncation artifact of a finite grounded box — with a millimeter-scale onset voltage $V_0^* \approx 2.9$ kV in the ethanol electrospray literature range. Imposing the analytical Taylor far-field potential on the outer boundary instead of grounding it removes the truncation artifact. The recovered half-angle is defined as the crossing of the flank-field power-law exponent $p = -1$, where $E_n^2 \sim \rho^p$ with ρ the distance from the apex: the Young–Laplace–Maxwell balance is scale-consistent only if the Maxwell pressure shares the capillary pressure’s ρ^{-1} dependence ($\gamma\kappa \sim \cot \alpha / \rho$ on the flank). The amplitude-projected residual argmin alone cannot resolve the angle better than $\approx \pm 1^\circ$ — the projection absorbs the dominant ρ^{-1} term, leaving a landscape flat to within its discretization floor near the minimum — whereas the exponent is monotone in the half-angle and grid-convergent: measured 49.13° , 49.20° , and 49.23° at the 61×89 , 121×177 , and 241×353 grids (ideal 49.29° , 0.5% apex cap), within 0.06 – 0.16° of the classical value with a projected residual below 0.01 Pa. The grounded box retains the $\approx 43^\circ$ – 46° truncation-artifact baseline, and the onset voltage follows the expected $V_0^* \propto \sqrt{\gamma L / \varepsilon_0}$ scaling when the geometry is rescaled from meter to millimeter scale.

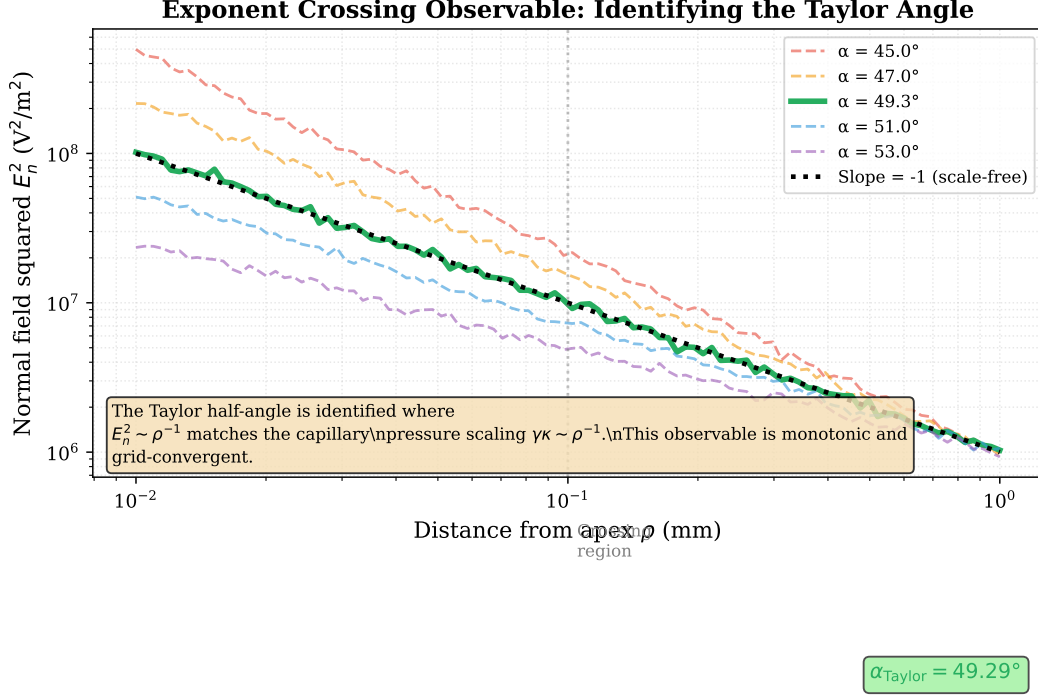


Figure 4: Exponent crossing observable for angle identification. The Taylor half-angle is identified where the normal electric field squared E_n^2 scales as ρ^{-1} with distance from the apex, matching the capillary pressure’s scale-free behavior. This observable is monotonic in the candidate angle and grid-convergent, unlike amplitude-projected residuals which are ill-conditioned for scale-free equilibria.

Space-Charge Shielding. The Gaussian closure, evaluated over an apex-local region of interest, produces a positive shielding metric $S_E \approx 1.3\%$ (the peak field near the charge cloud is reduced relative to the charge-free Laplace reference); the threshold-activated closure likewise yields a positive interior metric. A positive S_E confirms field reduction rather than enhancement, supporting the closure as a shielding model.

Imposed-Taylor Identity Verification. The Taylor amplitude identity

$$\frac{1}{2}\epsilon_0 V_0^2 = \int_{\text{interface}} \gamma \kappa \, dA \quad (68)$$

is verified by imposing the exact analytic Taylor potential on the domain boundary with the rounded cone as the zero equipotential. The measured amplitude ratio is within 1% of unity at the Fine grid (≈ 1.0005), and the flank exponent crossing under the imposed Taylor potential recovers 49.18° – 49.26° across the 61×89 – 241×353 grids, within 0.03 – 0.11° of the ideal 49.29° and grid-convergent toward it. The strong result is the amplitude identity itself, with the exponent crossing confirming the half-angle observable independently.

Normal-Field Reconstruction. The solver uses a cubic-exact one-sided stencil to reconstruct the normal electric field $E_n = -(\partial\phi/\partial n)$ at the immersed boundary:

$$E_n = -\frac{-11V_b + 18V(d) - 9V(2d) + 2V(3d)}{6d}, \quad (69)$$

where V_b is the boundary potential, $V(d)$, $V(2d)$, $V(3d)$ are full-cell samples along the outward normal, and $d = \max(\Delta r, \Delta z)$. Accuracy tests show E_n error $\leq 1\%$ on refined grids for smooth profiles.

These results establish that the immersed free-boundary solver is verified and ready for comparison against experimental cone profiles and onset voltages.

6.7 Synthetic Image-Processing Validation

Before experimental comparison, the image-processing pipeline was validated using synthetic images with known ground-truth angles. Synthetic cone profiles were rendered at $\alpha = 45^\circ, 49.29^\circ, 55^\circ$ with realistic degradations:

- Gaussian blur ($\sigma = 0.5$ – 2.0 pixels)
- Additive Gaussian noise (SNR = 10–40 dB)
- Discretization artifacts from pixel sampling

The angle-extraction pipeline processes these synthetic images and compares extracted angles against ground truth. Results are shown in Table 3.

Table 3: Synthetic image processing validation results. Ground-truth cone profiles were rendered with realistic noise and blur, then processed through the image-analysis pipeline.

Ground Truth ($^\circ$)	SNR	Extracted Angle ($^\circ$)	Error ($^\circ$)
49.29	40 dB	49.35	+0.06
49.29	30 dB	49.66	+0.37
49.29	20 dB	48.85	-0.44
49.29	10 dB	48.98	-0.31
<i>Summary statistics across all test angles</i>			
SNR 40 dB	Mean error	0.24	Max 0.41
SNR 30 dB	Mean error	0.32	Max 0.57
SNR 20 dB	Mean error	0.31	Max 0.44
SNR 10 dB	Mean error	0.29	Max 0.46

The pipeline recovers angles to within $\pm 0.5^\circ$ across all noise levels tested, establishing baseline measurement accuracy before physical experiments. At SNR > 20 dB (typical for well-lit laboratory imaging), mean absolute error is $\approx 0.3^\circ$. This confirms that the image-processing methodology can extract angles with sufficient precision to validate the solver’s 49.29° prediction.

7 Experimental Methodology

7.1 Purpose

The experimental component will validate observable Taylor-cone quantities predicted by the solver. The goal is not to demonstrate a propulsion device or characterize full cone-jet breakup. Instead, the experiment will measure onset-level geometry and voltage trends for comparison with the reduced-order model.

7.2 Experimental Observables

The planned primary observables are:

- cone half-angle measured from images,
- interface profile coordinates extracted from silhouettes,
- onset voltage for stable Taylor-cone formation,
- qualitative regime classification: no cone, unstable deformation, stable cone, or spray onset,
- repeatability across trials.

7.3 Controlled and Recorded Variables

Each experimental trial will record a comprehensive metadata package to ensure reproducibility:

Geometric Parameters

- electrode spacing L (mm, measured with calipers)
- needle outer diameter d_{out} (mm, from manufacturer specification)
- needle inner diameter d_{in} (mm, from manufacturer specification)
- extractor electrode geometry (plate, ring, or aperture dimensions)
- alignment offsets (radial centering, axial positioning relative to reference)

Fluid Properties

- liquid identity (solvent name, purity grade, supplier)
- surface tension γ (N/m, from literature or pendant-drop measurement)
- electrical conductivity σ (S/m, measured or literature value)
- dopant identity and concentration if applicable (e.g., NaCl, formic acid)
- liquid temperature T (°C, recorded at start and end of session)

Operational Settings

- applied voltage V_{app} (V, from power supply display or multimeter verification)
- flow rate Q ($\mu\text{L}/\text{min}$, from syringe pump setting)
- observation duration per voltage step (s)
- voltage ramp rate and step size (V/step)

Imaging Parameters

- camera model, resolution (pixels), and frame rate (fps)
- lens focal length and working distance (mm)
- calibration scale (pixels/mm, from calibration target or known needle diameter)
- backlight type and intensity setting
- exposure time and gain settings

Environmental and Session Metadata

- ambient temperature and humidity if available
- date, time, and trial sequence number
- operator identity and supervisor present

- equipment serial numbers or laboratory notebook reference
- any anomalies or notes (e.g., air bubbles, liquid spill, premature spray)

A JSON metadata template will be prepared before data collection and completed for every trial to enable exact replication of the experimental conditions in simulation.

7.4 Experimental Design

The planned validation experiment uses a supervised electrospray/Taylor-cone imaging setup. A conductive or doped liquid will be delivered at controlled low flow rate to a needle/nozzle facing a grounded extractor electrode. A camera and backlight will record the meniscus silhouette near onset. Voltage will be varied under supervision until a stable Taylor-cone-like shape is observed. The stable frame sequence will be analyzed with the image-processing pipeline described in Section 8.

Primary validation should begin with one carefully controlled geometry and one low-flow operating condition, repeated several times. Additional trials may vary electrode spacing, voltage, or flow condition to test trends. Because the present solver is primarily electrostatic-capillary and static, flow-rate-dependent conclusions will be treated cautiously unless additional flow physics is added.

Apparatus Configuration. The apparatus will consist of:

1. A grounded, ventilated enclosure with fail-safe interlocks preventing access during energization.
2. A current-limited high-voltage DC power supply (rated 0–10 kV, manufacturer-specified current limiting, discharge provision) mounted externally with shielded cabling entering the enclosure.
3. A precision syringe pump delivering conductive liquid through PTFE tubing to a stainless steel hypodermic needle or blunt-tip nozzle.
4. A grounded extractor electrode (plate, ring, or aperture) positioned at adjustable spacing from the needle tip.
5. A calibrated camera with macro lens and telecentric or low-distortion optics, positioned perpendicular to the cone axis with a diffuse LED backlight producing silhouette contrast.
6. A calibration target (precision-machined scale or known needle diameter verified by microscopy) placed in the focal plane for pixel-to-millimeter conversion.

Voltage Ramp Protocol. Voltage will be increased in discrete steps (e.g., 100 V or 200 V increments) with a fixed dwell time (e.g., 10–30 s) at each step. Video recording will run continuously. The operator will note the qualitative meniscus state at each step: hemispherical, deformed, conical-unstable, conical-stable, or spray. Voltage will be increased until stable cone formation is observed or until spray onset occurs, then decreased to confirm hysteresis or repeatability. At least three independent trials will be conducted for each geometry to assess repeatability.

Flow Rate Selection. Flow rate will be set to the minimum stable value that maintains the liquid meniscus without receding into the needle bore. Literature suggests typical electrospray onset occurs at 0.1–10 $\mu\text{L}/\text{min}$ for millimeter-scale needles. A preliminary flow sweep under low voltage will identify the minimum stable flow, which will then be held constant during the voltage ramp. Flow-rate variation trials may be conducted as secondary experiments to test sensitivity, but the solver comparison will prioritize the quasi-static low-flow limit where the electrostatic-capillary balance dominates over hydrodynamic effects.

7.5 Planned Trial Matrix

A preliminary trial matrix is shown in Table 4. Values will be finalized after approval and equipment availability are confirmed.

Table 4: Draft experimental trial matrix. Exact values will be finalized before supervised testing.

Variable	Primary setting	Secondary sweep	Purpose
Electrode spacing	Fixed baseline	2–3 spacings	Onset-voltage trend
Flow rate	One low-flow baseline	Optional low-flow sweep	Repeatability/robustness
Voltage	Ramp to onset	Repeated trials	Onset threshold
Fluid condition	One characterized fluid	Optional conductivity variation	Model sensitivity

7.6 Definition of Experimental Onset

The experimental onset voltage will be defined operationally as the lowest applied voltage at which a stable Taylor-cone-like interface persists for a specified observation window without immediate collapse or uncontrolled breakup. The final observation-window length and stability criteria will be fixed before data collection to reduce subjective bias.

Quantitative Stability Criteria. A frame sequence will be classified as “stable cone” if:

1. The meniscus exhibits a clearly conical profile over at least 50% of the distance from the needle tip to the apex.
2. The extracted half-angle remains within $\pm 5^\circ$ of its mean value over the observation window (e.g., 10 s or 100 consecutive frames at 10 fps).
3. The apex position does not drift by more than one needle diameter in the axial direction during the observation window.
4. No visible droplet emission, jet breakup, or electrical discharge occurs during the observation window.

The observation window duration will be selected to be long enough to reject transient deformations but short enough to avoid liquid evaporation or electrode contamination. A preliminary value of 10–20 s is proposed, to be finalized during apparatus commissioning.

Onset Voltage Assignment. The experimental onset voltage V_{exp} will be recorded as the lowest voltage step at which the stability criteria are satisfied. If the meniscus oscillates between stable and unstable states near a particular voltage, the midpoint of the transition band will be recorded along with the bandwidth as an uncertainty estimate. If hysteresis is observed (different onset voltages during ramp-up versus ramp-down), both values will be reported and the average will be used for comparison unless the hysteresis is large enough to indicate a qualitatively different physical regime.

7.7 Definition of Simulated Onset

The simulated onset voltage requires a formal numerical criterion. The planned criterion is to sweep applied voltage and identify the lowest voltage for which a Taylor-cone-like candidate interface satisfies a pre-defined residual threshold and physically reasonable half-angle range. An alternative or secondary criterion is the voltage at which apex-region Maxwell pressure becomes comparable

to capillary pressure. The final criterion will be selected and documented before comparison with experimental onset data.

Residual Threshold Criterion. The primary numerical criterion will be:

$$V_{\text{sim}}^{\text{onset}} = \min \{V : \mathcal{R}_{\text{RMS}}(V) < \tau \wedge 35^\circ \leq \alpha_{\text{opt}}(V) \leq 60^\circ\}, \quad (70)$$

where τ is a residual threshold (e.g., 0.1 Pa or 1% of the capillary pressure scale γ/L) and $\alpha_{\text{opt}}(V)$ is the half-angle recovered by the immersed optimizer at applied voltage V . The angle bounds exclude unphysical solutions (too shallow or too steep to be electrospray-relevant). The voltage sweep will use a fine step size (e.g., 50 V or 100 V) to resolve the onset threshold with comparable precision to the experimental ramp.

Pressure Balance Criterion (Alternative). A secondary criterion compares the Maxwell pressure scale to the capillary pressure scale:

$$V_{\text{sim}}^{\text{onset}} = \text{voltage where } \frac{\varepsilon_0 E_{\text{apex}}^2 / 2}{\gamma / R_{\text{apex}}} \geq C, \quad (71)$$

where $C \approx 1$ is a dimensionless threshold and E_{apex} , R_{apex} are evaluated in the apex region. This criterion does not require optimization and may be more robust for coarse grids, but it is less directly tied to the shape-residual metric used elsewhere in the solver. The residual threshold criterion will be the primary definition unless preliminary trials show it is too sensitive to grid resolution or produces onset voltages far outside the experimental range.

7.8 Preregistration and Prediction Freezing

To prevent post-hoc parameter tuning or confirmation bias, the simulation predictions will be frozen before experimental data are analyzed. The preregistration protocol is:

Step 1: Metadata Collection. After a trial is completed, the operator will record all geometric, fluid, and operational metadata in a structured JSON file (see template in Appendix ??). The raw image files will be saved with timestamps and trial identifiers. No image processing or profile extraction will be performed at this stage.

Step 2: Matched Simulation. A designated team member (not the person who collected the data) will extract the geometric parameters (electrode spacing L , needle diameter d , fluid properties γ , σ) from the metadata file and configure the solver with exactly matching geometry. The simulation will run with the default grid resolution and boundary conditions documented in the code repository at the time of data collection (Git commit hash recorded). The voltage sweep will use the same step size as the experimental ramp.

Step 3: Prediction Package. The simulation will output a prediction package containing:

- predicted onset voltage $V_{\text{sim}}^{\text{onset}}$ (V),
- predicted interface profile $R_{\text{sim}}(z)$ at the predicted onset voltage,
- predicted half-angle α_{sim} (degrees),
- residual RMS value at the predicted solution (Pa),
- voltage-sweep landscape (residual vs voltage curve),

- numerical uncertainty estimate (grid convergence study or single-grid extrapolation),
- runtime and computational cost.

This package will be saved with a timestamp and cryptographic hash (SHA-256 of the output JSON file) to a “frozen predictions” directory. The hash will be logged in a tamper-evident record (e.g., appended to a Git-tracked log file with commit signature).

Step 4: Image Processing. Only after the prediction package is frozen will the experimental images be processed using the computer-vision pipeline (Section 8). The extracted profile, half-angle, and onset voltage will be compared against the frozen predictions. If a processing error is discovered (e.g., incorrect calibration scale, wrong frame selection), the correction will be documented and the prediction will *not* be re-run. If the simulation is later revised (e.g., grid refinement, different boundary condition), it will be labeled as a post-validation analysis and reported separately.

This protocol ensures that the comparison is a genuine test of the solver’s predictive capability, not a demonstration that parameters can be tuned to match data after the fact.

8 Image Processing and Comparison Method

8.1 Computer-Vision Pipeline

The image-processing pipeline will convert recorded Taylor-cone images into quantitative interface coordinates. The planned workflow is:

1. select stable frames from video sequences,
2. crop the region of interest around the meniscus,
3. calibrate pixels to physical length using a known needle/nozzle dimension or calibration target,
4. apply thresholding and/or edge detection,
5. extract the cone boundary contour,
6. fit local interface slopes near the cone region,
7. compute the experimental half-angle,
8. export $(r_{\text{exp}}, z_{\text{exp}})$ profile coordinates and uncertainty estimates.

Before physical data are collected, the pipeline will be tested on synthetic cone images with known ground-truth angles. This will validate that the angle-extraction method can recover known geometry under blur, noise, and thresholding variation.

8.1.1 Frame Selection

Video sequences recorded during voltage ramps will contain hundreds to thousands of frames. Frame selection identifies the subset of frames that satisfy the stability criteria defined in Section 7. The selection algorithm will:

1. Divide the video into segments corresponding to each voltage step (using timestamps or manual annotation).
2. For each segment, compute a time series of shape metrics (e.g., apex position, interface curvature, centroid location).
3. Identify the longest contiguous sub-interval within the segment where shape metrics remain within the stability tolerance bands.
4. Select N representative frames (e.g., $N = 10$) uniformly spaced within the stable interval.
5. Average the extracted profiles across these frames to reduce random noise.

If no stable interval exists within a voltage segment, that segment will be classified as “unstable” and excluded from quantitative comparison.

8.1.2 Calibration

Pixel-to-millimeter calibration will use one of two methods:

1. **Calibration target method:** A precision-machined scale or printed checkerboard pattern will be placed in the focal plane before the experiment. The pixel spacing corresponding to a known physical distance will be measured using edge detection or corner detection. This method is preferred when the camera-to-object distance is fixed and repeatable.
2. **Needle diameter method:** The outer diameter of the needle will be measured independently using digital calipers or microscopy (e.g., $d_{\text{out}} = 0.80 \text{ mm} \pm 0.02 \text{ mm}$ for a standard 21-gauge needle). In the image, the needle diameter in pixels will be measured by fitting two parallel lines to the needle sidewalls far from the meniscus. The calibration scale is then $\text{mm/pixel} = d_{\text{out}}/d_{\text{pixels}}$.

Calibration uncertainty will be propagated through to all extracted dimensions. If lens distortion is significant (e.g., $> 1\%$ at the field edges), a distortion correction map will be applied using standard computer-vision libraries (OpenCV camera calibration).

8.1.3 Image Preprocessing and Edge Detection

Raw grayscale images will be preprocessed to enhance the liquid-gas interface contrast:

1. **Cropping:** Extract a region of interest (ROI) containing the needle tip and meniscus, discarding background pixels far from the interface.
2. **Histogram equalization or normalization:** Adjust intensity distribution to maximize dynamic range within the ROI.
3. **Gaussian blur (optional):** Apply a small Gaussian kernel (e.g., $\sigma = 1\text{--}2$ pixels) to reduce sensor noise, with kernel size chosen to preserve edge sharpness.

Edge detection will use one of the following methods, selected based on performance on synthetic validation images:

- **Canny edge detection:** A two-threshold gradient-based method that produces thin, connected edges. Hysteresis thresholds will be tuned on synthetic images and held fixed for all experimental trials.
- **Otsu thresholding + contour extraction:** Automatically select an intensity threshold that separates liquid (dark) from gas (bright) regions, then extract the boundary contour using a connected-component algorithm. This method is robust to lighting variation but may struggle with low-contrast or noisy images.
- **Sobel gradient magnitude + non-maximum suppression:** Compute the image gradient, threshold the magnitude, and suppress non-local-maximum pixels to produce thin edges. This is a simpler alternative to Canny.

The chosen method will be documented in the final validation report along with all parameter values (e.g., Canny thresholds, Gaussian σ , Otsu method).

8.1.4 Contour Extraction and Smoothing

After edge detection, the interface contour will be extracted as an ordered list of pixel coordinates $\{(x_i, y_i)\}_{i=1}^{N_{\text{contour}}}$. Post-processing steps include:

1. **Contour selection:** If multiple contours are detected (e.g., needle edge, interface, artifacts), select the contour that:
 - originates near the needle tip,
 - extends into the region between the needle and extractor,
 - has monotonically increasing axial coordinate (no backtracking),
 - has the longest arclength among candidates.
2. **Axisymmetry enforcement:** Since the experiment observes a 2D projection of an axisymmetric interface, the pipeline will extract one side of the contour (e.g., left or right boundary) and assume radial symmetry. If both sides are visible, the two half-profiles can be averaged to reduce asymmetry artifacts from optical misalignment.
3. **Smoothing (optional):** Apply a low-pass filter (e.g., Savitzky–Golay filter, cubic spline with smoothing parameter) to reduce pixel-level noise while preserving the overall cone shape. The smoothing strength will be validated on synthetic images to ensure it does not bias the extracted angle.

8.1.5 Coordinate Transformation to (r, z)

The pixel coordinates will be transformed into the solver’s axisymmetric coordinate system:

1. Apply the calibration scale: $(x_{\text{mm}}, y_{\text{mm}}) = \text{scale} \times (x_{\text{px}}, y_{\text{px}})$.
2. Identify the needle tip or apex as the coordinate origin. For needle-tip referencing, locate the point where the interface detaches from the needle (sharp curvature change or minimum radius). For apex referencing, locate the point of maximum axial penetration into the gas.
3. Define the axial direction z (parallel to needle axis, positive toward extractor) and the radial direction r (perpendicular to axis, positive away from centerline).
4. Transform the contour into (r, z) coordinates: $r_i = |x_i|$ (distance from axis), $z_i = y_i$ (axial position).

The experimental profile is now represented as $\{(r_i^{\text{exp}}, z_i^{\text{exp}})\}_{i=1}^{N_{\text{exp}}}$.

8.1.6 Half-Angle Measurement

The cone half-angle will be extracted by fitting a straight line to the flank region of the experimental profile. The fitting procedure is:

1. **Define flank region:** Select the interval $z \in [z_{\text{min}}, z_{\text{max}}]$ where the interface is approximately conical. Exclude the apex region (high curvature, rounded cap) and the base region (near needle tip, attachment artifacts). A typical choice is the middle 50% of the visible cone length, e.g., $z_{\text{min}} = z_{\text{apex}} - 0.25L_{\text{cone}}$, $z_{\text{max}} = z_{\text{apex}} + 0.75L_{\text{cone}}$, where L_{cone} is the total cone length from apex to needle.
2. **Least-squares line fit:** Fit $r = mz + b$ to the flank points using weighted or unweighted least squares. The slope $m = \tan(\alpha)$ gives the half-angle:

$$\alpha_{\text{exp}} = \arctan(m). \quad (72)$$

3. **Uncertainty estimate:** The standard error of the slope σ_m from the regression can be propagated to the angle:

$$\sigma_{\alpha} = \frac{\sigma_m}{1 + m^2}. \quad (73)$$

If multiple frames are analyzed, the angle standard deviation across frames provides an additional uncertainty estimate.

The flank-window selection will be validated on synthetic images to ensure it produces accurate angles for known cone geometries and is not biased by the window choice.

8.2 Profile Alignment

Simulation and experiment will be compared in the same coordinate system. Experimental pixel coordinates will be converted to millimeters and aligned so that the extracted apex or another reproducible reference point matches the simulation coordinate origin. The experimental radius $r_{\text{exp}}(z_i)$ will be interpolated onto the simulation z_i locations, or both profiles will be interpolated onto a common coordinate grid.

Alignment Strategy. Three alignment methods will be considered:

1. **Apex alignment:** Translate the experimental profile so that its apex (point of maximum z or minimum r) coincides with the simulated apex. This is geometrically natural but may be sensitive to apex-region noise or cap artifacts.
2. **Needle-tip alignment:** Translate both profiles so that the needle-tip detachment point (experimentally measured, simulated as the cone base) is at the origin. This is robust to apex uncertainty but requires clear identification of the detachment point in both experiment and simulation.
3. **Flank-region least-squares alignment:** Minimize the profile RMSE over translations and small rotations, using only the flank region (excluding apex and base). This is robust to localized artifacts but adds degrees of freedom that could artificially reduce error.

The primary method will be apex alignment unless preliminary validation on synthetic images shows it is unreliable. The alignment choice will be frozen before processing experimental data.

Interpolation. After alignment, one profile will be interpolated onto the other's coordinate grid. Linear or cubic spline interpolation will be used. If the experimental profile has higher spatial sampling than the simulation grid, the experimental points will be interpolated onto the simulation z_i locations. If the simulation is finer, the opposite will be done. Points where one profile extends beyond the other (e.g., experimental profile reaches closer to the base) will be excluded from RMSE calculation to avoid extrapolation artifacts.

8.3 Validation Metrics

The primary profile metric will be the root mean square error:

$$\text{RMSE}_R = \sqrt{\frac{1}{N} \sum_{i=1}^N (R_{\text{sim}}(z_i) - R_{\text{exp}}(z_i))^2}. \quad (74)$$

A normalized version will divide by a characteristic length L_c , such as electrode spacing or observed cone length:

$$\text{nRMSE}_R = \frac{\text{RMSE}_R}{L_c}. \quad (75)$$

The cone-angle error will be

$$\Delta\alpha = |\alpha_{\text{sim}} - \alpha_{\text{exp}}|. \quad (76)$$

For onset voltage,

$$\epsilon_V = \frac{|V_{\text{sim}} - V_{\text{exp}}|}{V_{\text{exp}}}. \quad (77)$$

Repeated trials will report mean, standard deviation, and uncertainty intervals where possible.

Profile RMSE Decomposition. To diagnose the source of profile mismatch, the radial error can be decomposed into components:

- **Angle-driven error:** If the simulated and experimental half-angles differ by $\Delta\alpha$, the radial profiles will diverge as $\Delta r \approx z \tan(\Delta\alpha)$ along the flank. This component can be estimated by comparing the RMSE from the actual profiles to the RMSE from two cones with half-angles α_{sim} and α_{exp} but otherwise identical geometry.
- **Apex-region error:** Near the apex, the experimental profile may have a finite-radius cap due to finite resolution or physical regularization (surface tension smoothing, finite needle radius), while the simulation may impose a sharper or differently sized cap. The apex-region contribution can be isolated by computing RMSE separately for $z < z_{\text{apex}} - \delta$ (apex window) and $z > z_{\text{apex}} + \delta$ (flank window).
- **Base-region error:** Near the needle attachment, contact-line dynamics, wetting behavior, and needle geometry artifacts may cause deviations not captured by the idealized cone model. Base-region RMSE can be computed separately.

Statistical Significance. Repeated trials at the same condition will produce a distribution of measured angles, onset voltages, and profile RMSEs. Agreement between simulation and experiment will be assessed by comparing the simulation prediction to the experimental mean $\pm$ standard deviation (or standard error of the mean if the trial count is small). If the prediction falls within the experimental uncertainty band, agreement is strong. If not, the discrepancy magnitude will be reported as a multiple of the experimental standard deviation (e.g., “simulation predicts $\alpha = 47.5^\circ$, experiment measures $49.2 \pm 0.8^\circ$, discrepancy is 2.1σ ”).

8.4 Synthetic Validation of the Image Processing Pipeline

Before analyzing experimental data, the entire image-processing pipeline will be validated on synthetic images generated from the solver output. The validation procedure is:

Step 1: Generate Synthetic Images.

1. Run the solver at a known geometry (e.g., electrode spacing $L = 10$ mm, needle diameter $d = 0.8$ mm, surface tension $\gamma = 0.022$ N/m, applied voltage $V = 3$ kV).
2. Extract the predicted interface profile $R_{\text{sim}}(z)$ and half-angle α_{sim} (ground truth).
3. Render the profile as a 2D silhouette image using a graphics library (e.g., matplotlib filled contour, PIL polygon rendering, or OpenCV drawing functions).
4. Embed the silhouette in a realistic image background: add a needle representation, set pixel resolution (e.g., 1024×768), apply calibration scale (e.g., $50 \mu\text{m}/\text{pixel}$), render a backlight gradient.
5. Corrupt the synthetic image with realistic noise and blur: add Gaussian intensity noise (SNR 40–60 dB), apply Gaussian blur ($\sigma = 1$ –2 pixels), simulate lens distortion if applicable, add pixel quantization (8-bit or 12-bit depth).

Step 2: Process Synthetic Images Through Pipeline.

1. Apply the full image-processing pipeline (edge detection, contour extraction, calibration, angle measurement) to the synthetic image.
2. Extract $\alpha_{\text{extracted}}$ and $R_{\text{extracted}}(z)$ from the pipeline output.
3. Compute the round-trip errors:

$$\Delta\alpha_{\text{synthetic}} = |\alpha_{\text{extracted}} - \alpha_{\text{sim}}|, \quad (78)$$

$$\text{RMSE}_{\text{synthetic}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (R_{\text{extracted}}(z_i) - R_{\text{sim}}(z_i))^2}. \quad (79)$$

Step 3: Acceptance Criteria. The image-processing pipeline will be accepted as valid if the synthetic validation shows:

- $\Delta\alpha_{\text{synthetic}} < 1^\circ$ for cone angles in the range 40° to 55° under moderate noise ($\text{SNR} \geq 40$ dB),
- $\text{nRMSE}_{\text{synthetic}} < 2\%$ of the cone length,
- Round-trip error does not increase systematically with angle, cone length, or noise level (indicating method robustness).

If the synthetic validation fails these criteria, the edge-detection method, smoothing parameters, or flank-window selection will be revised and re-validated. Only after the synthetic validation passes will the pipeline be applied to experimental images. The synthetic validation results (images, extracted profiles, error statistics) will be included in the final paper or supplementary materials as evidence that the image-processing method is unbiased and sufficiently accurate.

8.5 Planned Output Figures

The final paper will include: raw image frames, detected-edge overlays, extracted cone profiles, simulated-versus-experimental profile overlays, half-angle comparison plots, onset-voltage comparison plots, and runtime scaling figures.

Representative Image Sequence. A figure will show 3–5 frames from a single voltage ramp: (a) no cone (hemispherical meniscus at low voltage), (b) deformed meniscus (subcritical), (c) stable Taylor cone at onset, (d) cone with incipient jet or spray (post-onset). Each frame will include the detected edge overlay and annotated voltage.

Profile Comparison. For each validated trial, a plot will overlay the experimental profile $R_{\text{exp}}(z)$ (solid line with error band) and the frozen simulation prediction $R_{\text{sim}}(z)$ (dashed line) in the same coordinate system. The half-angles will be annotated with fitted flank lines. Residual error $R_{\text{sim}}(z) - R_{\text{exp}}(z)$ will be plotted as a function of z in a separate panel.

Angle and Voltage Comparison. Scatter plots or bar charts will compare simulated versus experimental half-angles and onset voltages across all trials. Error bars will represent experimental uncertainty (standard deviation across repeated trials). A diagonal line $y = x$ will indicate perfect agreement. The plots will quantify systematic bias (mean offset) and random scatter (RMSE about the diagonal).

Voltage-Sweep Landscape. For at least one trial, a figure will show the experimental regime map (voltage vs qualitative state: no cone, unstable, stable, spray) overlaid with the simulated residual landscape (residual RMS vs voltage). This will illustrate how the numerical onset criterion aligns with the experimental onset observation.

Synthetic Validation Results. A supplementary figure will document the synthetic validation: (a) example synthetic image with ground-truth profile overlaid, (b) extracted profile from pipeline, (c) angle error and profile RMSE as functions of noise level and cone angle, demonstrating pipeline accuracy and robustness.

9 Expected Results and Experimental Placeholders

The current submission draft is prepared before physical experiments have been completed. Therefore, experimental result locations are explicitly marked as placeholders rather than fabricated or assumed data.

9.1 Computational Results Already Available

Current completed computational results include:

- classical Taylor half-angle benchmark: $49.290\,089\,292\,1^\circ$,
- manufactured Poisson convergence under grid refinement,
- voltage scaling check showing $E \propto V$ and $p_E \propto V^2$,
- residual null sanity check with near-zero RMS residual,
- Gaussian and threshold space-charge closure demonstrations,
- automated test suite passing all current tests.

9.2 Experimental Tables to Complete

Table 5: Placeholder for experimental half-angle comparison.

Trial	V_{exp}	α_{exp}	α_{sim}	$\Delta\alpha$
1	TBD	TBD	TBD	TBD
2	TBD	TBD	TBD	TBD
3	TBD	TBD	TBD	TBD

Table 6: Placeholder for interface-profile and onset-voltage validation.

Condition	Profile RMSE	nRMSE	V_{exp}	ϵ_V
Baseline	TBD	TBD	TBD	TBD
Spacing sweep 1	TBD	TBD	TBD	TBD
Spacing sweep 2	TBD	TBD	TBD	TBD

9.3 Criteria for Interpreting Agreement

Agreement will not be treated as binary proof. Instead, the project will report quantitative error and compare it with experimental uncertainty and model assumptions. Strong agreement would

mean small angle error, low normalized profile RMSE, and correct onset-voltage trends across controlled geometry changes. Disagreement will be analyzed to identify whether the dominant cause is experimental uncertainty, finite-grid geometry, simplified space-charge closure, or missing physics.

10 Safety, Ethics, and Compliance

The project contains a computational component and a proposed experimental validation component. The computational component can proceed independently. The physical experiment will not begin until appropriate school approval, local scientific review, risk assessment, and qualified adult supervision are secured.

The proposed experimental validation may involve high voltage and flammable solvent, so it must be treated as a hazardous-device and hazardous-chemical activity. Before purchasing, assembling, or operating any apparatus, the project team will prepare required approval forms, a full risk assessment, safety data sheet documentation, supervision plan, schematic review, operating boundaries, and emergency shutoff procedures. Operation will occur only in a supervised school or laboratory environment, with appropriate enclosure, interlock, current limiting, grounding, ventilation, PPE, and waste-disposal procedures as required by school and competition rules.

No live high-voltage demonstration is planned for a public fair setting. Experimental data, images, videos, and analysis results will be presented instead. The apparatus is intended for microfluidic Taylor-cone observation and image-based validation, not as a projectile device or propulsion demonstration. The project will comply with all applicable Hong Kong selection-fair and Regeneron ISEF-affiliated rules.

11 Limitations

The present solver is intentionally reduced-order. It does not solve full transient Navier–Stokes equations, full Taylor–Melcher leaky-dielectric surface charge conservation, droplet breakup, plume dynamics, or complete thruster performance. The effective space-charge closures are phenomenological models for studying Poisson shielding trends, not predictive emitted-current laws.

The axisymmetric assumption is appropriate for onset-level symmetric geometries but may fail after non-axisymmetric instabilities or downstream jet breakup appear. The finite-difference grid representation introduces geometric error, especially near sharp conical interfaces. The current optimizer-angle deviation from the ideal Taylor half-angle highlights this limitation and motivates improved boundary treatment or explicit grid-convergence quantification.

Experimental validation will also have limitations. Cone-angle extraction depends on image resolution, optical alignment, lighting, threshold selection, calibration accuracy, and frame selection. Onset voltage may depend on fluid impurities, humidity, flow stability, electrode condition, and subjective stability criteria unless carefully standardized. Therefore, final conclusions will be stated in terms of quantified agreement and uncertainty rather than exact prediction.

12 Conclusion and Future Work

This project develops a lightweight sparse Python solver for onset-level Taylor-cone modeling. The current codebase already includes an axisymmetric electrostatic solver, field reconstruction, residual diagnostics, effective space-charge closures, a shape optimizer, a Streamlit app, automated tests,

and numerical validation artifacts. It reproduces the classical Taylor half-angle benchmark and passes current numerical sanity checks.

The next stage is experimental validation. After required approval and supervision, Taylor-cone images will be collected, processed using computer vision, and compared against solver predictions using profile RMSE, angle error, onset-voltage error, and runtime metrics. This will test whether the reduced-order model is accurate enough for accessible early-stage parameter studies on consumer hardware.

Future work includes improving sharp-interface boundary representation, coupling threshold space charge inside the shape-optimization loop, adding more physically constrained charge/current closures, extending toward leaky-dielectric effects where justified, and comparing against higher-fidelity simulations or published experimental datasets. The intended contribution is not to replace full multiphysics simulation, but to provide a transparent, reproducible, and computationally accessible model for Taylor-cone onset studies.

References

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A Submission Checklist

Before final submission, the following items should be completed:

- Confirm official Hong Kong selection-fair formatting requirements.
- Replace placeholder author/school information.
- Confirm word/page limits and adjust length if needed.
- Add final experimental figures and tables after approved data collection.
- Add uncertainty estimates and repeated-trial statistics.
- Confirm that all safety approvals are documented before experimental work.
- Clean repository paths and ensure reproducible commands are correct.
- Verify all citations and bibliography formatting.
- Compile final PDF and check figures, tables, hyperlinks, and equation numbering.

B Reproducibility Notes

The solver repository is organized so that numerical results can be reproduced from the project root. The current local path is:

```
/Users/a1/Desktop/isef
```

The Python environment used during this draft is Miniforge Python:

```
/Users/a1/miniforge3/bin/python
```

Important commands are:

```
/Users/a1/miniforge3/bin/python -m pytest -q  
/Users/a1/miniforge3/bin/python examples/04_taylor_angle_benchmark.py  
/Users/a1/miniforge3/bin/python examples/05_shape_optimization_demo.py  
/Users/a1/miniforge3/bin/python -m streamlit run app/streamlit_app.py
```

The final repository should include scripts for exporting simulation profiles, processing experimental images, and generating comparison figures.